AI CONCEPTS:

1. LLM (Large Language Model)

What?

A large AI model trained on massive text data.

Example

- ChatGPT
- Gemini
- Claude

Architecture

User
↓
Prompt
↓
LLM
↓
Generated Response

Example:

User: Explain Java OOPs
↓
GPT
↓
Answer

---

2. Tokens

What?

LLMs don't read words directly.

They break text into small pieces called tokens.

Example

Java Developer

becomes

Java
Developer

Architecture

Input Text
↓
Tokenizer
↓
Tokens
↓
LLM

---

3. Context Window

What?

How much information the LLM can remember in one conversation.

Example

Question 1
Question 2
Question 3

LLM remembers previous questions.

Architecture

Previous Chats
+
Current Prompt
↓
Context
↓
LLM

---

4. Prompt Engineering

What?

Writing effective instructions for AI.

Example

Bad:

Teach Java

Good:

Act as a Java Instructor.
Explain OOPs with examples.

Architecture

Prompt
↓
LLM
↓
Better Output

---

5. Embeddings

What?

Convert text into numbers.

Example

Java Developer

↓

[0.25,0.87,0.43...]

Architecture

Text
↓
Embedding Model
↓
Vector

---

6. Vector Database

What?

Stores embeddings.

Popular Tools

- Pinecone
- ChromaDB
- FAISS
- Weaviate

Architecture

Documents
↓
Embeddings
↓
Vector Database

---

7. Semantic Search

What?

Search by meaning instead of keywords.

Example

Search:

How to center div?

Finds:

CSS Alignment Guide

Even without exact words.

Architecture

Question
↓
Embedding
↓
Vector Search
↓
Similar Results

---

8. RAG (Retrieval Augmented Generation)

Most Important

Problem

LLM doesn't know company data.

Solution

Retrieve documents before answering.

Example

Company Chatbot

Architecture

User Question
↓
Create Embedding
↓
Vector Database
↓
Relevant Documents
↓
LLM
↓
Answer

Example:

Employee:
Leave Policy?

↓

Company Documents

↓

LLM

↓

Answer

---

9. AI Agent

What?

AI that can plan and execute tasks.

Example

Book Flight
Book Hotel
Send Email

without human intervention.

Architecture

Goal
↓
Agent
↓
Think
↓
Act
↓
Result

---

10. Tool Calling

What?

Agent uses external tools.

Example

Calculator
Weather API
Database
Email

Architecture

User Question
↓
Agent
↓
Choose Tool
↓
Tool Execution
↓
Answer

---

11. Agent Memory

What?

Agent remembers previous conversations.

Example

User:
My name is Mohan

Later:

What is my name?

Agent remembers.

Architecture

User
↓
Memory Store
↓
Agent
↓
Response

---

12. LangChain

What?

Framework for building AI applications.

Features

- RAG
- Agents
- Memory
- Tools

Architecture

User
↓
LangChain
↓
LLM
↓
Tools
↓
Response

---

13. LangGraph

What?

Framework for stateful AI agents.

Why?

Agents can make decisions.

Architecture

Start
↓
Search
↓
Think
↓
Decision
↙ ↘
Yes No
↓ ↓
Action Retry

---

14. MCP (Model Context Protocol)

What?

Standard protocol connecting AI with tools.

Easy Analogy

USB-C for AI

Example

Connect AI to:

- GitHub
- Database
- Files
- Slack

Architecture

AI Assistant
↓
MCP Client
↓
MCP Server
↓
GitHub/File/DB

Without MCP:

AI
↓
Custom Integration
↓
Tool

With MCP:

AI
↓
MCP
↓
Any MCP Tool

---

15. Multi-Agent Systems

What?

Multiple agents working together.

Example

Building a software project.

Agent 1

Research Agent

Agent 2

Developer Agent

Agent 3

Tester Agent

Architecture

User Request
↓
Manager Agent
↙ ↓ ↘
Research Dev Tester
↘ ↓ ↙
Result

Complete Modern AI Architecture

This is the architecture used in many current AI products:

User
↓
Prompt
↓
AI Agent
↓
Memory
↓
Tool Calling
↓
MCP
↓
External Tools
↓
RAG
↓
Vector Database
↓
Documents
↓
LLM
↓
Final Response

Interview Priority Order

Learn these in order:

1. LLM
2. Tokens
3. Context Window
4. Prompt Engineering
5. Embeddings
6. Vector Database
7. Semantic Search
8. RAG
9. AI Agents
10. Tool Calling
11. Agent Memory
12. LangChain
13. LangGraph
14. MCP
15. Multi-Agent Systems

Interview Questions:

If you're preparing for **modern AI interviews**, you should know the AI concepts that are currently used in real-world AI applications and AI agent systems.

Level 1: AI Fundamentals

1. Machine Learning (ML)

What?

- Computers learn patterns from data.

Example

- Spam email detection.

---

2. Deep Learning (DL)

What?

- ML using neural networks.

Example

- Face recognition.

---

3. Generative AI

What?

- AI that generates content.

Example

- ChatGPT generating text.
- Midjourney generating images.

Level 2: LLM Concepts

4. LLM (Large Language Model)

What?

- A huge neural network trained on massive text data.

Examples

- OpenAI GPT
- Google Gemini
- Anthropic Claude
- Meta Llama

---

5. Tokens

What?

- Small text pieces processed by LLMs.

Example:

Hello World

Might become:

Hello
World

---

6. Context Window

What?

- Amount of information an LLM can remember in one conversation.

---

7. Prompt Engineering

What?

- Writing effective instructions for AI.

Example:

Act as a Java Interviewer.
Ask 5 DSA questions.

---

8. Hallucination

What?

- AI generates incorrect information confidently.

---

9. Fine Tuning

What?

- Training an LLM further on specific data.

Example:

- Hospital AI
- Banking AI

Level 3: RAG Stack

10. Embeddings

What?

- Numerical representation of text.

Example:

Java Developer

↓

[0.12,0.67,0.34...]

---

11. Vector Database

Purpose

- Stores embeddings.

Examples:

- ChromaDB
- Pinecone
- FAISS
- Weaviate

---

12. Semantic Search

What?

- Search based on meaning rather than keywords.

Example:

How to center div?

Finds:

CSS alignment guide

---

13. RAG (Retrieval Augmented Generation)

Most Important Interview Topic

Flow

User Question
↓
Vector Search
↓
Relevant Documents
↓
LLM
↓
Answer

Example:

- Company chatbot using internal documents.

Level 4: AI Agents

14. AI Agent

What?

- AI that can think, plan, and act.

Example:

Book flight
Find hotel
Create itinerary

Without human intervention.

---

15. Tool Calling

What?

- Agent can use tools.

Examples:

Calculator
Database
API
Email
Browser

---

16. Agent Memory

What?

- Agent remembers previous interactions.

---

17. Planning

What?

- Agent breaks a task into steps.

Example:

Build website
↓
Create backend
↓
Create frontend
↓
Deploy

---

18. Multi-Agent Systems

What?

- Multiple agents collaborate.

Example:

Research Agent
Coding Agent
Testing Agent

Level 5: Modern Agent Frameworks

19. LangChain

Most popular framework.

Used for:

- RAG
- Agents
- Tool calling

---

20. LangGraph

Very important now.

Why?

Builds stateful AI agents.

Example:

Agent
↓
Search
↓
Think
↓
Decide
↓
Act

---

21. CrewAI

Creates teams of AI agents.

Example:

Manager Agent
Developer Agent
Tester Agent

---

22. AutoGen

Framework for multi-agent communication.

Level 6: MCP (Very Hot Topic)

23. MCP (Model Context Protocol)

What?

A standard that allows AI models to connect with external tools and data sources.

Think of MCP as:

USB-C for AI

Just like USB-C lets devices connect universally,

MCP lets AI connect universally to:

- Databases
- GitHub
- Files
- APIs
- Business tools

Why MCP?

Without MCP:

LLM (AI Models)
↓
Custom Integration
↓
Tool

For every tool separately.

With MCP:

LLM (AI Models)
↓
MCP
↓
Any MCP Tool

One standard for all.

MCP Components

Host

Example:

- AI Assistant

Client

Communicates with MCP servers.

MCP Server

Provides tools/data.

Example:

GitHub MCP Server
Database MCP Server
File MCP Server

Level 7: AI System Design Concepts

24. Function Calling

LLM decides which function to execute.

---

25. Tool Use

AI uses external systems.

---

26. Workflow Orchestration

Managing AI task execution.

---

27. Guardrails

Safety controls.

Example:

Block harmful output

---

28. Human-in-the-Loop

Human approval before critical actions.

---

29. Evaluation (Evals)

Testing AI quality.

---

30. AI Observability

Monitoring AI systems.

Top 15 Must-Know Concepts for Interviews

1. LLM
2. Tokens
3. Context Window
4. Prompt Engineering
5. Embeddings
6. Vector Databases
7. Semantic Search
8. RAG
9. AI Agents
10. Tool Calling
11. Agent Memory
12. LangChain
13. LangGraph
14. MCP
15. Multi-Agent Systems